PROFESSOR
FERRETTO

Física

ESTÁTICA

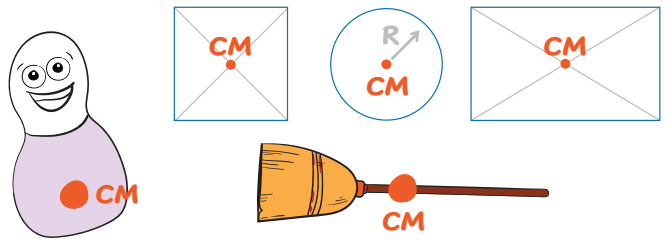
FÍSICA

PROFESSOR COELHO

ESTÁTICA

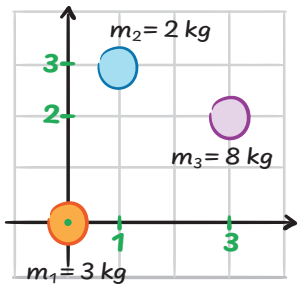
Centro de Massa (CM)

Em corpos regulares, o centro de massa é localizado no centro geométrico.



Em corpos irregulares, o centro de massa é mais próximo da região de maior massa.

Determinação analítica

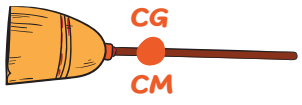


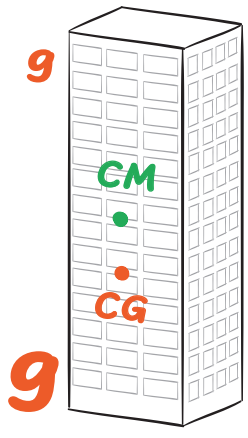
$$x_{CM} = \frac{x_1 \cdot m_1 + x_2 \cdot m_2 + x_3 \cdot m_3}{m_{TOTAL}}$$

$$y_{CM} = \frac{y_1 \cdot m_1 + y_2 \cdot m_2 + y_3 \cdot m_3}{m_{TOTAL}}$$

Centro de Gravidade (CG)

Ponto onde pode-se imaginar a **aplicação da força peso.**



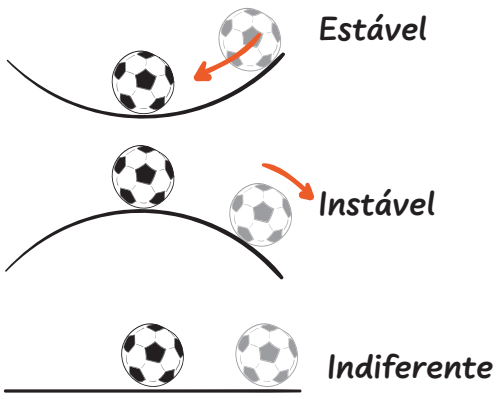


Em campos gravitacionais uniformes, o CG coincide com o CM.

Em corpos de altura considerável, o CG é deslocado para baixo do CM.

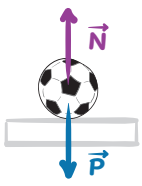
EQUILÍBRIO DE UM PONTO MATERIAL

Tipos de Equilíbrio



Condição de Equilíbrio

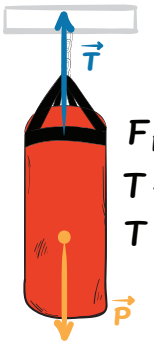
$F_R = 0$ ou $\Sigma \vec{F} = 0$



$F_R = 0$

$N - P = 0$

$N = P$

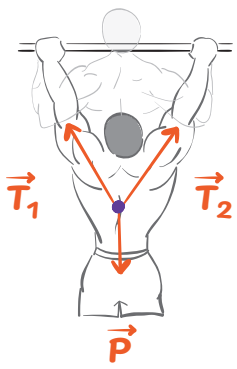


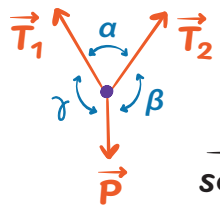
$F_R = 0$

$T - P = 0$

$T = P$

Teorema de Lamy (somente para 3 forças)

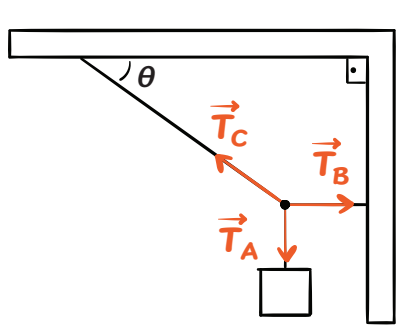


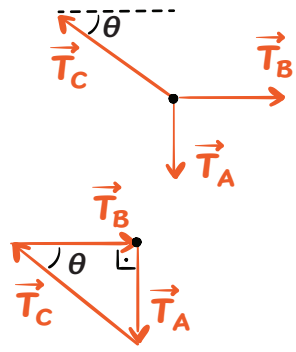


$$\frac{P}{sen(\alpha)} = \frac{T_1}{sen(\beta)} = \frac{T_2}{sen(\gamma)}$$

Esta equação também é conhecida como **Lei dos Senos.**

Método do Polígono

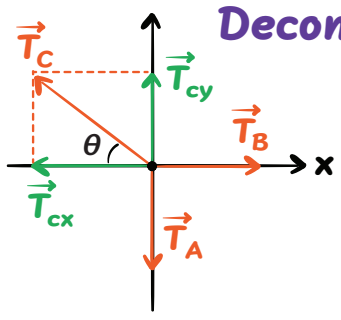




$$sen(\theta) = \frac{T_A}{T_C}$$

$$cos(\theta) = \frac{T_B}{T_C}$$

Decomposição Vetorial

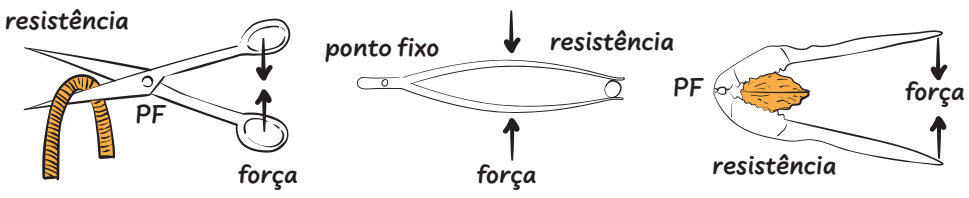


$T_{cx} = T_C \cdot cos(\theta)$ $\quad \Sigma \vec{F}_X = 0$

$T_{cy} = T_C \cdot sen(\theta)$ $\quad \Sigma \vec{F}_Y = 0$

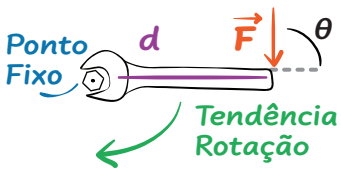
EQUILÍBRIO DE UM CORPO EXTENSO

Alavancas



Interfixa | Interpotente | Inter-resistente

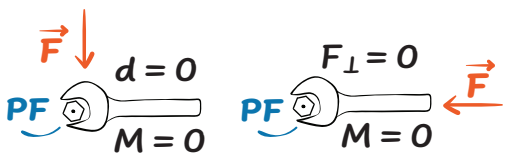
Momento de uma força (torque)



$M = F \cdot sen(\theta) \cdot d$

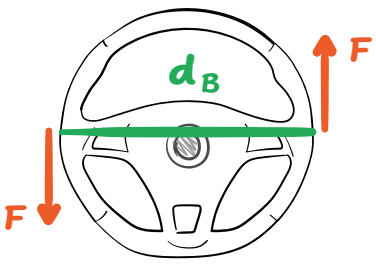
$M = F_\perp \cdot d$

Força perpendicular ao braço de ação.



Convenção de sinais

Momento Binário

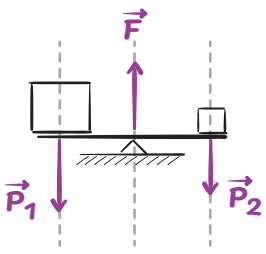


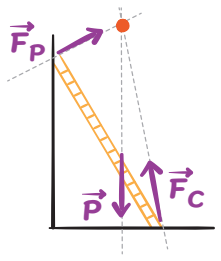
$M_B = F \cdot d_B$

As forças são paralelas e de sentidos opostos. O momento binário independe da localização do ponto fixo.

Condições de Equilíbrio

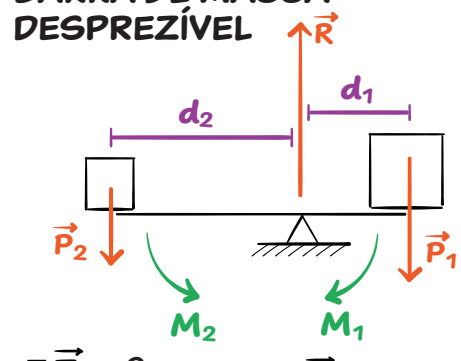
$\Sigma \vec{F} = 0$ $\quad \Sigma \vec{M} = 0$





Se um corpo extenso está em equilíbrio pela ação de apenas três forças, elas devem ser paralelas ou concorrentes.

BARRA DE MASSA DESPREZÍVEL



$\Sigma \vec{F} = 0$

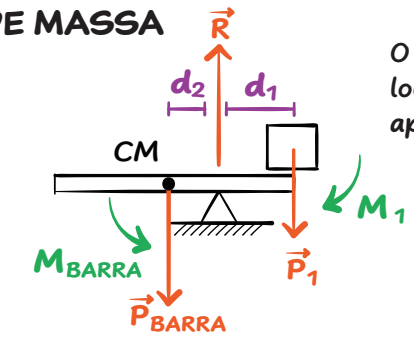
$R = P_1 + P_2$

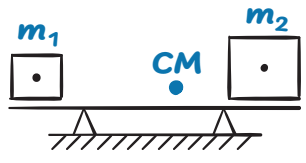
$\Sigma \vec{M} = 0$

$M_1 = M_2$

$P_1 \cdot d_1 = P_2 \cdot d_2$

BARRA DOTADA DE MASSA



Basta localizar o CM da barra e aplicar seu peso ali.

O CM do sistema deve estar localizado entre os pontos de apoio ou sobre algum deles.